

EE236 Lab 10 Report

C-V Characteristics of MOS Capacitor

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1 Aim of the experiment

The aim of this experiment is to -

- Create a summer and amplifier circuit and test its working.
- Find the variation of C_{DUT} with V_{DC} and calculate the MOSCAP parameters - Oxide capacitance, Oxide thickness, Doping density, Flat band voltage, Flat band capacitance, Debye length, Debye capacitance.

2 Summer and Amplifier Circuit

2.1 Circuit Diagram

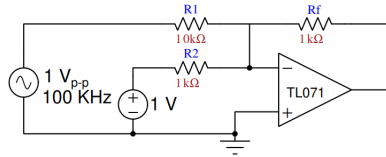


Figure 1: Summer Circuit

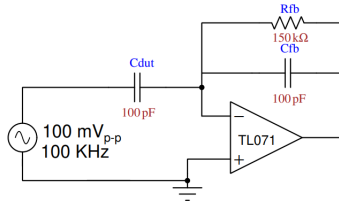


Figure 2: Amplifier Circuit

2.2 Explanations and Experimental Results

2.2.1 Summer Circuit

The circuit returns the weighted sum of inputs correctly. The DC voltage is measured using a DMM and the output waveform from the oscilloscope.

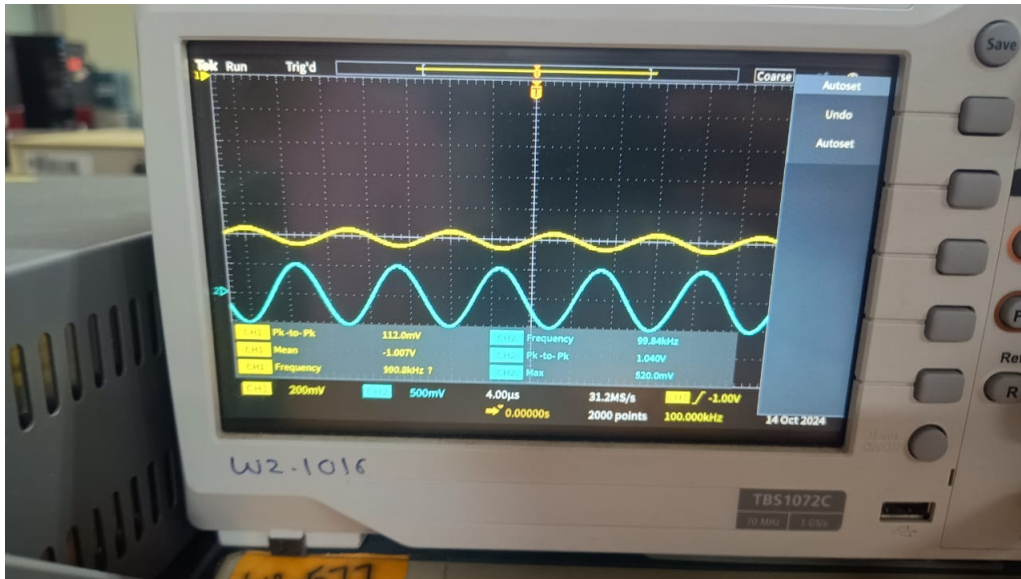


Figure 3: Summer Circuit Output

2.2.2 Amplifier Circuit

The circuit works well at higher frequencies but starts to give shaky outputs when the frequency decreases beyond 5kHz. It acts as a high pass filter.

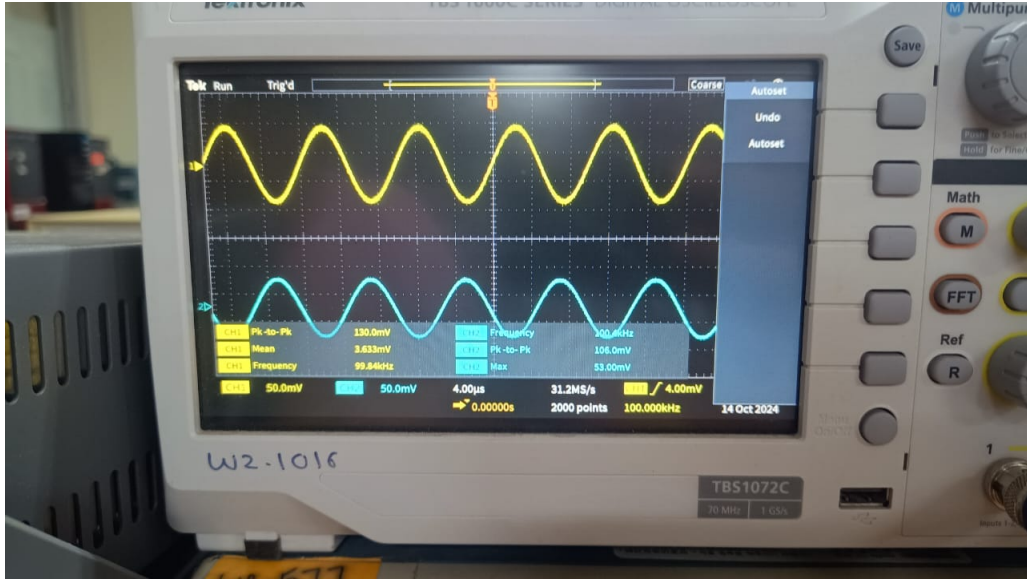


Figure 4: Amplifier Circuit Output

3 Measurement of MOSCAP C-V characteristics

3.1 Circuit Diagram

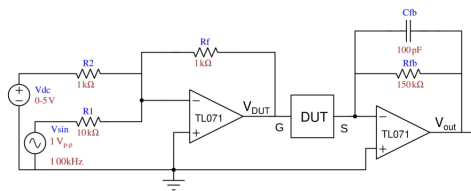


Figure 5: MOSCAP Circuit

3.2 Experimental Results

3.2.1 C_{DUT} vs V_{DC}

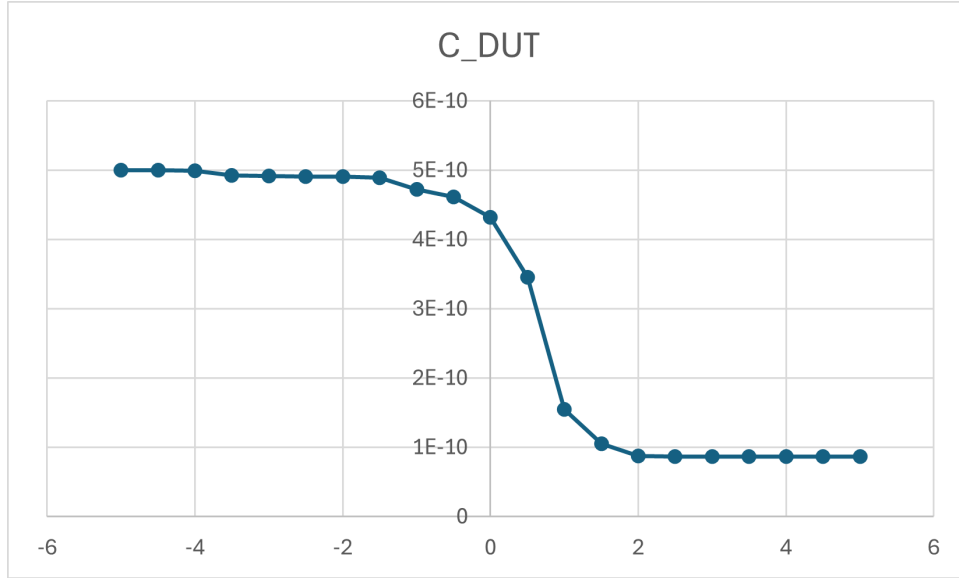


Figure 6: PCAP

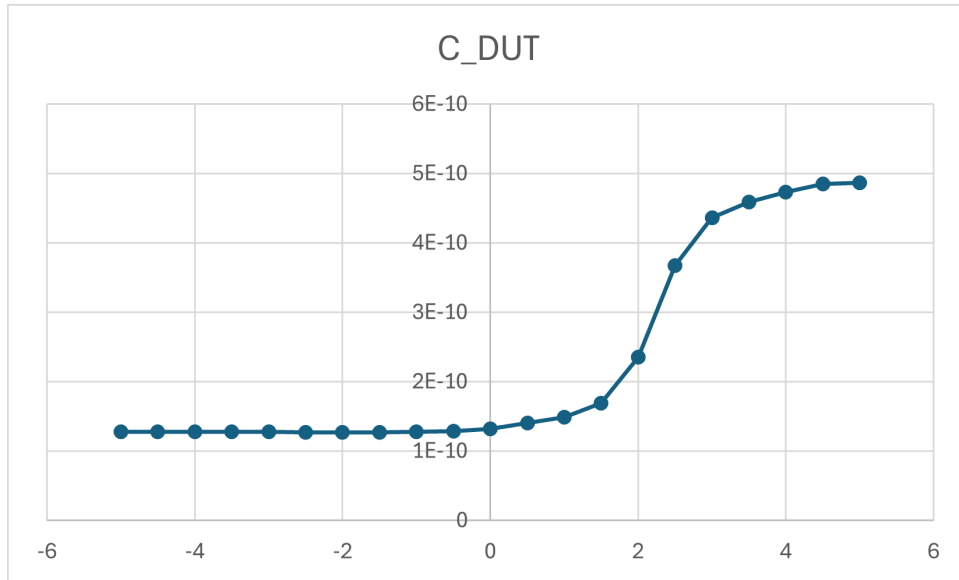


Figure 7: NCAP

V_{DC} is the input voltage to the circuit, the bias voltage applied to MOSCAP is actually $-V_{DC}$.

3.3 Calculation of MOSCAP Parameters (for NCAP)

3.3.1 Oxide Capacitance

From the graph of C_{DUT} vs V_{DC} , maximum capacitance is C_{ox} .

$$C_{ox} = 487pF$$

3.3.2 Oxide Thickness

From the above value of C_{ox} , the value of oxide thickness is:

$$t_{ox} = 70.87nm$$

3.3.3 Doping Density

The relation between C_s , C_{ox} , and C_{min} is:

$$C_{min} = \frac{C_{ox}C_s}{C_{ox} + C_s}$$

$$C_s = \frac{C_{min}C_{ox}}{C_{ox} - C_{min}}$$

Substituting the values ($C_{min} = 128pF$):

$$C_s = 1.735 \times 10^{-10} F/cm^2$$

$$t_{dep} = \frac{\varepsilon_{si}}{C_s}$$

Using the permittivity of silicon ε_{si} ,

$$\varepsilon_{si} = 11.7 \times \varepsilon_0 = 1.035 \times 10^{-12} F/cm$$

The value of t_{dep} is:

$$t_{dep} = \frac{1.035 \times 10^{-12}}{1.735 \times 10^{-10}} = 59.7 \text{ nm}$$

$$t_{dep} = \sqrt{\frac{2\varepsilon_{si}}{qN_A} \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right)}$$

Where $q = 1.6 \times 10^{-19} C$, $kT = 0.0259 \text{ eV}$ at room temperature, $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.
Using an iterative solver, the value of N_A is:

$$N_A = 2.27 \times 10^{10} /cm^3$$

3.3.4 Debye Length

$$L_{\text{debye}} = \sqrt{\frac{\varepsilon_{si} kT}{q^2 N_A}}$$

$$L_{\text{debye}} = \sqrt{\frac{(11.7)(8.854 \times 10^{-14})(0.026 \text{ eV} \times 1.6 \times 10^{-19} \text{ J/eV})}{(1.6 \times 10^{-19})^2 (2.27 \times 10^{10})}}$$

$$L_{\text{debye}} \approx 8.6 \times 10^{-4} \text{ cm}$$

3.3.5 Debye Capacitance

$$C_{\text{debye}} = \frac{\varepsilon_{si}}{L_{\text{debye}}}$$

$$C_{\text{debye}} = \frac{(11.7)(8.854 \times 10^{-14})}{8.6 \times 10^{-4}} \text{ F/cm}^2$$

$$C_{\text{debye}} \approx 1.204 \times 10^{-9} \text{ F/cm}^2$$

3.3.6 Flatband Capacitance

$$C_{\text{fb}} = \frac{C_{ox} \cdot C_{\text{debye}}}{C_{ox} + C_{\text{debye}}}$$

$$C_{\text{fb}} = \frac{(4.87 \times 10^{-10}) \cdot (1.204 \times 10^{-9})}{4.87 \times 10^{-10} + 1.204 \times 10^{-9}}$$

$$C_{\text{fb}} \approx 3.467 \times 10^{-10} \text{ F/cm}^2$$

3.3.7 Flatband Voltage

From the graph of C_{DUT} vs V_{DC} , the value of Flatband Voltage obtained is -

$$V_{fb} = 2.75 \text{ V}$$

3.4 Calculation of MOSCAP Parameters (for PCAP)

Going through the same method and calculations as NCAP, the following values of parameters are obtained -

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The values used from the graph in above calculations are -

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